G_2 Symmetry Breaking in the Universal Somatic Field: The Biological Emotional Attractor and Geometric Ideal

$W = W_{G_2} + \delta W$: Decomposing the BRECVEMA Coupling Matrix

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Abstract

The BRECVEMA coupling matrix W_8 of the Universal Somatic Field decomposes uniquely as $W_8 = W_{G_2} + \delta W$, where $W_{G_2} = \frac{6}{5} I_8$ is the G_2-invariant component and δW is a traceless symmetry-breaking term. The G_2-symmetric limit W_{G_2} corresponds to perfectly balanced emotional processing — all eight BRECVEMA mechanisms equally coupled, no directional anisotropy. The empirical biological matrix W_8 (calibrated from Juslin 2019) is 48.4% symmetry-broken from this ideal: $\|\delta W\|_F / \|W_8\|_F = 0.484$. The symmetry-breaking modes are traceless (their eigenvalues sum to zero) and correspond to specific emotional dynamics: strong positive anisotropy in the VI–EM and ME–AJ couplings; negative anisotropy in the BS–AJ channel (stress suppresses aesthetics). Therapeutic intervention reduces $\|\delta W\|_F$, driving the system toward the G_2 attractor. This provides the first quantitative connection between a G_2-symmetric compactification interpretation and the biological coupling matrix. It resolves the algebraic $8 \to 7$ reduction in the coupling decomposition; it does not derive a G_2-holonomy metric for X_7.

Contents

1

Introduction: The 8→7 Dimension Question

Dark Matter as the Spatial Vacuum of the Universal Somatic Field (Johnson, 2026) proposes that dark matter is the vacuum energy of the three non-compact spatial dimensions of the USF and derived the cosmological energy budget from the dimensional partition $11 = 7 + 3 + 1$. The compact sector In an M-theory compactification interpretation, a compact seven-manifold may carry G_2 holonomy. The current USF formal core proves only a well-defined flat 7D product for X_7. But the biological emotional field is 8-dimensional (BRECVEMA, eight mechanisms). How does its algebraic structure relate to a seven-dimensional compact-sector interpretation?

This paper resolves the question. The 8D BRECVEMA field W_8 decomposes as the G_2-invariant part plus a traceless symmetry-breaking term. The G_2-invariant part is exactly $\frac{6}{5}I_8$ — a diagonal matrix. The remaining 7 off-diagonal degrees of freedom constitute the symmetry-breaking δW, which lives in the 7D adjoint representation of G_2. The biological emotional system operates on the 8D field, while its traceless symmetry-breaking sector is seven-dimensional. This supplies an algebraic compatibility result for a 7D compact-sector interpretation; it does not by itself prove the metric geometry or holonomy of X_7.

2

The G₂-Symmetric Limit

The USF coupling matrix W_8 acts on the 8D BRECVEMA state space $\psi = (\psi_0, \dots, \psi_7)$ where the indices correspond to the eight mechanisms: BrainStem (BS), Rhythmic Entrainment (RE), Evaluative Conditioning (EC), Contagion (CO), Visual Imagery (VI), Episodic Memory (EM), Musical Expectancy (ME), Aesthetic Judgement (AJ).

Definition. A matrix W acting on $\mathbb{R}^8$ is G_2-invariant if it commutes with all G_2 transformations. By Schur's lemma, since $\mathbb{R}^8$ decomposes under G_2 as $\mathbb{R}^1 \oplus \mathbb{R}^7$ (real part $\oplus$ imaginary octonions), a G_2-invariant matrix must be block-diagonal: $W_{G_2} = \lambda_0 P_0 + \lambda_1 P_1$, where P_0, P_1 are projections onto the two invariant subspaces.

For the USF, the self-coupling sets $\lambda_0 = \lambda_1 = \frac{6}{5}$ (the diagonal of W_8), giving:

$$W_{G_2} = \frac{6}{5} I_8$$

This is the G_2-symmetric attractor: all eight mechanisms are equally coupled, no mechanism is privileged. In the G_2-symmetric limit, the emotional field has maximal symmetry — no directional anisotropy, no preferred emotional mode.

3

The Decomposition of W₈

The empirical matrix W_8 (calibrated from Juslin 2019, Table 22.3) has diagonal entries all equal to $\frac{6}{5}$ and non-zero off-diagonal entries:

Coupling	Value
$W_{BS,EC}$	$+3/10$

Coupling	Value
$W_{BS,CO}$	$+2/5$
$W_{RE,CO}$	$+1/2$
$W_{EC,CO}$	$+2/5$
$W_{VI,EM}$	$+3/5$
$W_{ME,AJ}$	$+7/10$
$W_{BS,AJ}$	$-2/5$ (negative)
$W_{EC,VI}$	$-3/10$ (negative)

The unique decomposition $W_8 = W_{G_2} + \delta W$ gives:

$$\delta W = W_8 - \frac{6}{5} I_8$$

δW is traceless by construction: $\mathrm{tr}(\delta W) = \mathrm{tr}(W_8) - 8 \cdot \frac{6}{5} = \frac{48}{5} - \frac{48}{5} = 0$.

Key numerical results:

$$\|\delta W\|_F = 1.876, \quad \|W_8\|_F = 3.877$$
$$\frac{\|\delta W\|_F}{\|W_8\|_F} = 0.484 \quad \text{(48.4\% symmetry broken)}$$

The eigenvalues of δW (sorted): $+0.984,\ +0.718,\ +0.591,\ +0.113,\ -0.226,\ -0.585,\ -0.742,\ -$

Their sum is exactly zero (tracelessness). The spectrum is non-degenerate: biological emotional processing is not G_2-symmetric at any sub-eigenspace level.

4

Physical Interpretation of the Symmetry-Breaking Modes

The traceless matrix δW encodes the biological anisotropies of emotional processing. Its non-zero entries correspond to:

Positive anisotropy (stronger coupling than the G_2 ideal): - $\delta W_{ME,AJ} = +0.7$: Musical expectancy strongly drives aesthetic judgment — the strongest anisotropy in the biological system - $\delta W_{VI,EM} = +0.6$: Visual imagery and episodic memory are tightly coupled - $\delta W_{RE,CO} = +0.5$: Rhythmic entrainment drives emotional contagion - $\delta W_{BS,CO} = +0.4$, $\delta W_{EC,CO} = +0.4$: Arousal and conditioning both activate social contagion

Negative anisotropy (weaker coupling than the G_2 ideal; anti-correlation): - $\delta W_{BS,AJ} = -0.4$: BrainStem arousal and Aesthetic Judgement are *anti-correlated* in the biological system — when physiological arousal is high, aesthetic appreciation is suppressed. This matches the known psychophysiology of stress and flow states. - $\delta W_{EC,VI} = -0.3$: Evaluative conditioning and visual imagery are anti-correlated — conditioned fear suppresses imagery (consistent with PTSD phenomenology)

The G_2 interpretation: The positive anisotropies represent the biological "short-cuts" — emotional couplings stronger than the symmetric ideal. The negative anisotropies represent the biological "blockers" — couplings weaker than symmetry would predict.

5

Therapeutic Trajectory: Reducing δW

The decomposition suggests a precise model of the therapeutic process:

Healthy processing corresponds to $\|\delta W\|_F \to 0$ — the emotional coupling approaching the G_2-symmetric ideal. Each coupling relaxes toward $\frac{6}{5}$: the strongest couplings weaken, the weakest strengthen, the negative couplings (BS–AJ, EC–VI) return to zero.

Trauma corresponds to a large $\|\delta W\|_F$ with specific anisotropies amplified: deep trauma strengthens the BS–AJ anti-correlation (arousal blocks aesthetic experience) and the EC–VI anti-correlation (conditioned responses block visual processing).

The somatic invariant: $\mathrm{tr}(\delta W) = 0$ is preserved throughout. This is the conservation law: the total energy of the symmetry-breaking modes is zero. No therapeutic intervention can add or remove total δW energy — it can only redistribute it. The goal of therapy is to drive δW toward a uniform distribution across all modes (which by tracelessness approaches zero entry-by-entry as the system approaches the G_2 attractor).

6

Formal Status

Statement	Lean location	Status
$W_{G_2} = (6/5)I_8$ defined	`BRECVEMAVariational.lean`	proved (`native_decide`)
$\delta W = W_8 - W_{G_2}$ traceless	`BRECVEMAVariational.lean`	proved (`norm_num`)
G_2-invariant matrix = λI_n	Schur's lemma	interpretation requires Lie-theory formalisation
moduli_space_is_G2_homotopy	`BRECVEMAVariational.lean`	open formalisation (gauge constraint derivation)

The numerical decomposition ($\|\delta W\|_F/\|W_8\|_F = 0.484$) is computed exactly using the rational matrix entries; Python floating-point is used only for display. The tracelessness of δW is an exact rational identity.

7

Conclusion

The 8-dimensional BRECVEMA coupling matrix W_8 decomposes uniquely as:

$$\boxed{W_8 = \frac{6}{5} I_8 + \delta W, \quad \mathrm{tr}(\delta W) = 0, \quad \|\delta W\|_F/\|W_8\|_F = 0.484}$$

The G_2-symmetric component $\frac{6}{5}I_8$ is the mathematical ideal of balanced emotional processing. The traceless symmetry-breaking δW encodes the biological anisotropies: the ME–AJ and VI–EM couplings are the dominant positive anisotropies; the BS–AJ anti-correlation (stress suppresses aesthetics) is the dominant negative anisotropy. Therapeutic progress corresponds to $\|\delta W\|_F \to 0$ while $\mathrm{tr}(\delta W) = 0$ is conserved.

The $8 \to 7$ algebraic reduction is resolved: the 8D biological field has a seven-dimensional traceless symmetry-breaking sector, compatible with the 7D compact-sector interpretation used by the companion cosmological-constant and spatial-vacuum papers. Deriving a compact G_2-holonomy metric remains a separate open geometric problem.

References

Johnson, A. (2026). *Dark matter as the spatial vacuum of the universal somatic field:* $\Omega_{\mathrm{DM}} = 3/11$.